# Fixed Income: Bond Basics and Valuation

LECTURE 10 · REVIEW NOTES
Luis Ceballos, PhD · FINA 402 · Fall 2026

We have been valuing equity by asking what future cash flows are worth today. With a bond, the idea is the same. What changes is the claim: a plain fixed-rate bond specifies coupon payments and a principal repayment date. I want you to read that contract first, then decide how to discount the payments.

$$P_0 = \frac{CF_1}{1+r} + \frac{CF_2}{(1+r)^2} + \cdots + \frac{CF_N}{(1+r)^N} = \sum_{k=1}^{N} \frac{CF_k}{(1+r)^k}$$

Here the discount rate is per payment period. A single rate is a useful starting point; later we will give each payment its own maturity-specific spot rate.

## 1. What does fixed income actually fix?

The contract fixes the promised payments for this type of bond. It does not fix the price at which you can sell tomorrow. And a promise is not a guarantee: default can change what investors actually receive. Our numerical exercises assume payments arrive in full and on time, with no fees or taxes.

## 2. Read the terms before calculating

| Term | What I want you to identify |
|---|---|
| Face value (FV or F) | Principal repaid at maturity; not necessarily the purchase price. |
| Coupon rate | Annual coupon dollars as a fraction of face value. |
| Payment frequency | How often coupons arrive; semiannual means twice a year. |
| Maturity | Years remaining until the final payment. |
| Spot rate | Discount rate for a cash flow at one specific maturity. |
| YTM | One internal rate of return that reproduces the observed bond price. |

## 3. The market is larger than government debt

Governments, corporations and households all borrow. In the lecture chart, Treasuries account for 60.5% of the selected $51.4 trillion debt categories, not 60.5% of all U.S. debt. That chart excludes mortgage-backed and asset-backed securities. Mortgage rates generally move with Treasury yields, but their spread can change. Keep the chart denominator in mind when interpreting any market share.

# From the contract to the cash-flow timeline

## 4. The Apple example in class

The Apple 2035 bond has a 4.750% annual coupon and pays twice a year. At its May 2025 issuance, the price was 99.340 per $100 face and the quoted YTM was 4.834%. These are historical issue terms, not a current market quote. We illustrate the cash flows per $1,000 face; the actual minimum denomination was $2,000.

$$C = \frac{F \times c}{m} = \frac{1{,}000 \times 0.0475}{2} = \$23.75$$

The investor pays $993.40 per $1,000 face. There are 20 semiannual coupons over ten years. The last payment includes both the final $23.75 coupon and the $1,000 principal. Do not use the purchase price to calculate the coupon.

$$CF_{20} = C + F = 23.75 + 1{,}000 = \$1{,}023.75$$

## 5. Our two-year teaching bond

For the remaining calculations, use $1,000 face value, a 6% annual coupon rate, two years to maturity and semiannual payments. This is a hypothetical teaching bond, distinct from the Apple issue.

$$C = \frac{1{,}000 \times 0.06}{2} = \$30, \qquad N = mT = 2 \times 2 = 4$$

| Maturity (years) | 0.5 | 1.0 | 1.5 | 2.0 |
|---|---|---|---|---|
| Cash flow ($) | 30 | 30 | 30 | 1,030 |

I use T for years and N for the number of payments throughout these notes. Discounting four six-month payments means four periods. It does not mean four years. The principal appears only in the last cash flow.

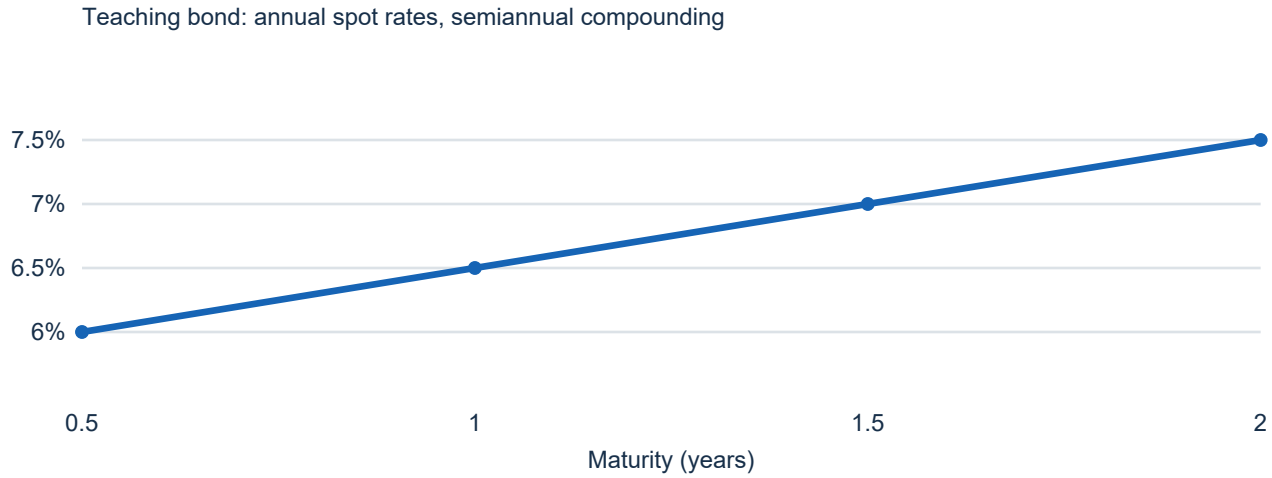
## 6. When one annual yield is given

$$P_0 = \sum_{k=1}^{N} \frac{C}{(1 + y/m)^k} + \frac{F}{(1 + y/m)^N}$$

If the quoted annual YTM is 8%, the six-month rate is 4%. Using the same cash flows gives a price of $963.70. This is the flat-yield illustration; it is a different discount-rate assumption from the spot-curve example on the next page.

# Price each payment with its matching spot rate

Now use the class example: annual spot rates of 6%, 6.5%, 7% and 7.5% for maturities of 0.5, 1, 1.5 and 2 years. These quotes use semiannual compounding. Each payment gets the rate for its own maturity.




$$DF_k = \frac{1}{(1 + s_k/2)^k}, \qquad P_0 = \sum_{k=1}^{4} CF_k \times DF_k$$

| Years | Spot rate | Cash flow | Discount factor | PV ($) |
|---|---|---|---|---|
| 0.5 | 6.00% | 30.00 | 0.970874 | 29.1262 |
| 1.0 | 6.50% | 30.00 | 0.938037 | 28.1411 |
| 1.5 | 7.00% | 30.00 | 0.901943 | 27.0583 |
| 2.0 | 7.50% | 1,030.00 | 0.863073 | 888.9653 |

$$P_0 = \frac{30}{1.03} + \frac{30}{1.0325^2} + \frac{30}{1.035^3} + \frac{1{,}030}{1.0375^4} = \boxed{\$973.2909}$$

The spot rate is already the rate appropriate to that maturity. Do not multiply successive spot rates together. For the final payment, divide 7.5% by two and raise the discounting term to four. Keep full precision in the calculation and round the final price.

## Use the Excel template in the same order

In the Example tab, the spot curve is already provided. In the Activity tab, first copy your own four assigned spot rates into B21:B24 as percentages, for example 6%. The shared workbook does not load your personal rates automatically. Fill the yellow B13:B16 cells for coupon, number of payments, price and YTM, using the suggested formulas alongside them. The maturity column reads 0.5, 1.0, 1.5 and 2.0 years.

# One YTM, the same price

Once the spot curve gives us a price, we can ask a different question: what single yield would produce that exact price? That is YTM. We solve for it; we do not take the arithmetic average of the spot rates.

$$973.290887 = \frac{30}{1+y/2} + \frac{30}{(1+y/2)^2} + \frac{30}{(1+y/2)^3} + \frac{1{,}030}{(1+y/2)^4}$$

$$y/2 = 3.731153\%, \qquad y = \boxed{7.462306\%}$$

In Excel, use **=2*RATE(4,30,-973.29088704,1000)** and format the result as a percentage. The price is a negative cash flow because you pay it today. Coupons and principal are positive receipts. RATE returns the six-month yield; multiplying by two gives the annual quoted YTM.

$$\text{Effective annual yield} = (1+y/2)^2 - 1 = 7.6015\%$$

The effective annual yield is a different convention. If the question asks for quoted annual YTM, report 7.4623%, not the effective yield. For a coupon bond, matching a realized compounded return to YTM also requires the relevant reinvestment assumption, as well as receiving the promised payments.

## Premium, par or discount?

| Coupon rate versus YTM | Price versus face | Classification |
|---|---|---|
| Coupon rate > YTM | P > FV | Premium |
| Coupon rate = YTM | P = FV | Par |
| Coupon rate < YTM | P < FV | Discount |

These comparisons apply to our plain fixed-rate bond valued on a coupon date. Here the coupon rate is 6%, below the 7.4623% YTM, and the price is below $1,000. The bond trades at a discount. Investors receive the fixed coupons plus a gain toward face value if the bond is held to maturity and paid as promised.

## The five activity outputs

Calculate the coupon payment, number of payments, spot-discounted price and quoted annual YTM, then classify the bond as premium, par or discount. Each part is worth one point. Everyone has the same face value, coupon rate and maturity; assigned spot rates vary, so your price and YTM can differ. The numerical answers allow a relative tolerance of 1%.

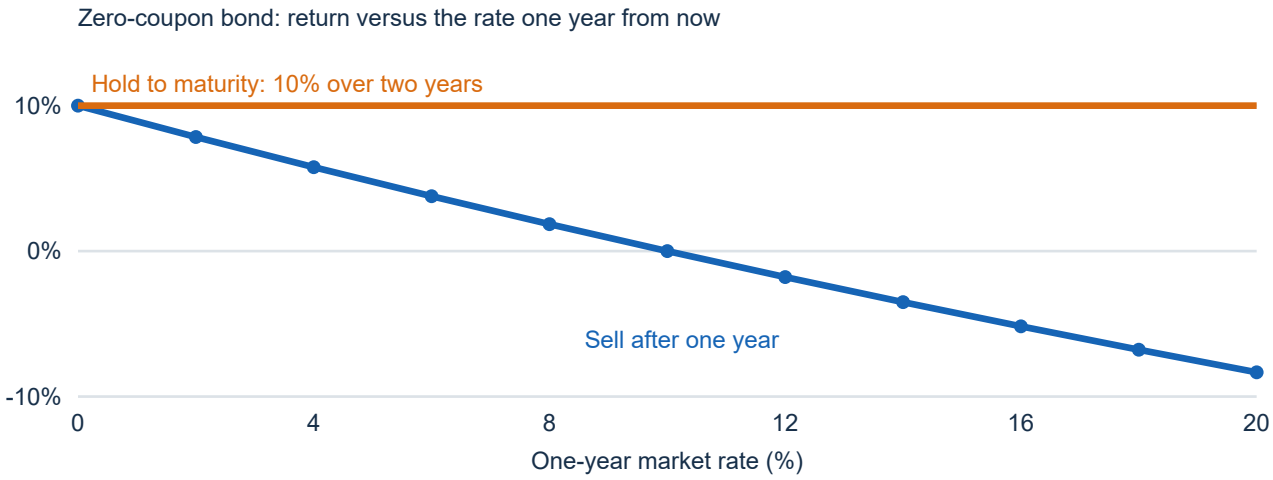
# A fixed payment is not a fixed resale return

The class survey uses a zero-coupon bond purchased for $1,000 today that promises $1,100 in two years. There are no intermediate coupons. Assume no default, taxes or transaction costs.

## Sell after one year

$$P_1 = \frac{1{,}100}{1 + r_1}, \qquad HPR_1 = \frac{P_1 - 1{,}000}{1{,}000}$$

At that point only one year remains. If the market rate is 5%, the resale price is $1,047.62 and the one-year return is 4.76%. If the rate is 15%, the resale price is $956.52 and the return is -4.35%. We know the final payment today, but not the future market rate or resale price.




## Hold to maturity

$$HPR_2 = \frac{1{,}100 - 1{,}000}{1{,}000} = 10\%, \qquad r_{\text{annual}} = \left(\frac{1{,}100}{1{,}000}\right)^{1/2} - 1 = 4.8809\%$$

The 10% is a total return over two years, not an annual return. The graph compares different holding horizons: the blue curve is a one-year sale return; the orange line is a two-year holding return. The two-year result is fixed under our assumptions because there are no interim coupons to reinvest and the final payment is fixed.

## What I would check before submitting

Start with the timeline. Apply the coupon rate to face value, match each spot rate to its cash flow, and keep payment frequency consistent. Calculate YTM from the price rather than averaging rates. Finally, tell me whether a return refers to selling early or holding to maturity, and whether it is annualized. Those distinctions matter more than memorizing a formula.